

Problem 10:

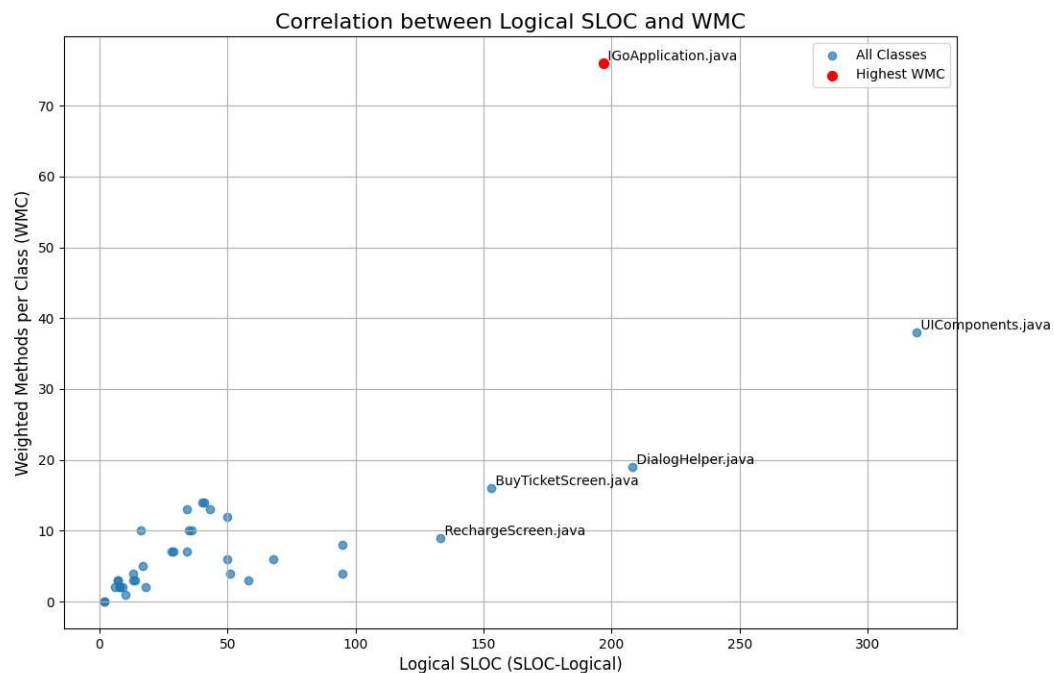
a) Scatter Plot Analysis:

1. Create scatter plot:

- X-axis: Logical SLOC for each class
- Y-axis: WMC for each class
- Plot each class as a point

2. Visual analysis:

- Look for linear relationship patterns
- Identify outliers
- Describe the trend (positive, negative, or no correlation)



Visual Analysis:

- **Trend:** The plot clearly shows a **strong positive linear relationship**. As the SLOC-Logical increases, the WMC also tends to increase. The points follow a pattern from the bottom-left corner (low SLOC, low WMC) to the top-right corner (high SLOC, high WMC).
- **Outliers/Key Points:** The class `iGoApplication.java` is highlighted in red. It stands out significantly from the main cluster, with by far the highest SLOC-Logical (291) and the highest WMC (83). This identifies it as the largest and most complex class in the system. The other classes are clustered more tightly in the lower-left, indicating they are generally smaller and less complex.

b) Correlation Coefficient:

1. Calculate Pearson correlation coefficient (r):

Recalculation of Correlation Coefficient (Part b)

Here are the new values for the 36 classes and the step-by-step calculation based on the formula:

$$r = \frac{\sum[(x_i - \bar{x})(y_i - \bar{y})]}{\sqrt{[\sum(x_i - \bar{x})^2 \times \sum(y_i - \bar{y})^2]}}$$

Step-by-Step Calculation:

1. Calculate the means ($\bar{x}$ and $\bar{y}$):

- n (Number of classes) = 36
- $\bar{x}$ (Mean of SLOC-Logical) = 54.0833
- $\bar{y}$ (Mean of WMC) = 9.3889

2. Calculate the Numerator ($\sum[(x_i - \bar{x})(y_i - \bar{y})]$):

- This is the sum of the product of deviations for each class.
- Numerator = 23246.8333

3. Calculate the Denominator Parts:

- Sum of squared deviations for X ($\sum(x_i - \bar{x})^2$): 166712.7500
- Sum of squared deviations for Y ($\sum(y_i - \bar{y})^2$): 6340.5556

4. Calculate the final r value:

- $r = 23246.8333 / \sqrt{[166712.7500 \times 6340.5556]}$
- $r = 23246.8333 / \sqrt{[1056910600]}$
- $r = 23246.8333 / 32510.1615$
- $r \approx 0.7150$

Updated Coefficient Summary & Insights

The recalculated Pearson correlation coefficient is $r = 0.7150$.

2. **Interpret r value:**
 - $|r| < 0.3$: Weak correlation
 - $0.3 \leq |r| < 0.7$: Moderate correlation
 - $|r| \geq 0.7$: Strong correlation
3. The calculated Pearson correlation coefficient is **$r = 0.7150$** . According to your interpretation scale ($|r| \geq 0.7$), this indicates a **strong positive correlation**. This quantitative result strongly reinforces the visual analysis from the scatter plot. It provides statistical evidence that in this project, larger classes (in terms of SLOC-Logical) are indeed proportionally more complex (in terms of WMC). While correlation does not equal causation, this strong link suggests that IGoApplication.java is not an anomaly but rather the extreme end of a clear trend. From a maintainability perspective, this metric correlation implies that any effort to reduce class size (SLOC) might be an effective strategy for also managing and reducing class complexity (WMC), which in turn improves testability and understandability.

Expected Relationship:

Your lecture notes mention that there's often a correlation between SLOC and complexity metrics like WMC, but they are **different attributes**. You should expect:

- Positive correlation (more code $\rightarrow$ more methods $\rightarrow$ higher WMC)
- But note: size $\neq$ complexity

Sensible Conclusions:

- Discuss whether larger classes (SLOC) have proportionally higher complexity (WMC)
- Mention outliers and what they might indicate
- Comment on maintainability implications
- Note limitations: correlation $\neq$ causation